

CIRCUITLENS2.0 – A SMALL-SCALE MECHANISTIC INTERPRETABILITY EXPERIMENT

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ABSTRACT. We investigate whether GPT-2 Small exhibits the characteristic behavior of induction heads, a well-studied mechanism for in-context sequence copying. We construct controlled sequences containing repeated token patterns and measure whether attention heads attend from a repeated token to the token that followed its earlier occurrence. We then use causal ablation to test whether the heads identified by this attention-based screen actually contribute to the behavior. Across 200 repeated sequences, the model exhibits a large induction effect: the mean target logit margin is 2.5717 for repeated sequences compared with 0.0150 for shuffled controls. The attention screen recovers several heads previously reported as induction heads in GPT-2 Small, with L5H5 obtaining the highest attention score. Causal ablation provides further evidence that several of these heads contribute positively to induction. At the same time, L10H7 provides an important counterexample: despite having a relatively high induction attention score, ablating it increases induction performance. This illustrates why attention patterns alone are insufficient for establishing a mechanistic explanation and motivates the use of causal interventions in mechanistic interpretability.

1. INTRODUCTION

Large language models are often treated as black boxes: given an input, they produce an output, but it is difficult to determine which internal computations produced that output. *Mechanistic interpretability* tries to open this box by identifying specific components of a neural network that implement recognizable computations and then testing those hypotheses using causal interventions. Rather than asking only whether a model behaves in a particular way, the goal is to understand how that behavior is implemented internally.

This distinction is important for understanding modern transformer language models. A model may contain billions of learned parameters, but its behavior can sometimes be explained in terms of smaller structures such as attention heads, neurons, and interactions between layers. If these structures can be identified and understood, mechanistic interpretability may provide a more systematic way of analyzing why models behave as they do and, ultimately, of investigating potential failure modes and safety-relevant behavior.

One of the best-studied examples of such an internal mechanism is the *induction head*, which implements a simple form of in-context copying (see Section 2 for a more detailed discussion). Induction heads are a useful and relatively concrete example of a mechanism through which a transformer can use information from earlier parts of its input [Ols+22].

Induction heads are also interesting because they provide a tractable case where an internal computation can be investigated at the level of individual attention heads. The mechanism can be described relatively simply, allowing us to move from a behavioral observation to an explicit hypothesis about what individual components of the model are doing.

This report presents a small, from-scratch investigation of this mechanism in GPT-2 Small. The experiment is deliberately designed to be simple enough to inspect and reproduce on a consumer computer while still following the basic logic of mechanistic interpretability research. The central question is:

How does GPT-2 Small implement induction and token-copying behavior?

The investigation proceeds in two stages. First, we screen all 144 attention heads in GPT-2 Small for the characteristic attention pattern expected from an induction head. Second, we intervene on the most promising candidates by ablating their outputs and measure the effect on induction performance.

The distinction between these two stages is essential. An attention pattern can show that a head is looking at the location we expect, but this does not by itself establish that the head is responsible for the model’s behavior. A head could be redundant, could be performing a different computation that happens to produce a similar pattern, or could even work against the behavior being measured. Causal ablation therefore provides a second, independent test.

The experiment produces three main findings. First, GPT-2 Small shows a clear induction effect on the controlled sequences used here. Second, the attention-based screen recovers several heads that closely match induction heads reported in previous work. Third, causal intervention reveals that the relationship between attention and function is not one-to-one: while several high-scoring heads contribute positively to induction, L10H7 has a relatively strong induction-like attention pattern but its ablation actually improves the measured behavior. This last result is particularly useful because it demonstrates, on a small scale, why mechanistic interpretability requires causal analysis rather than relying on correlations in activations alone.

2. BACKGROUND: INDUCTION HEADS

2.1. Transformer layers and attention heads. GPT-2 is a transformer language model. It processes a sequence of tokens through a stack of transformer layers, with each layer containing multiple attention heads operating in parallel. GPT-2 Small contains 12 transformer layers and 12 attention heads per layer, giving a total of 144 attention heads.

An attention head determines how information from different positions in the input sequence should be combined when computing the representation at a particular position. For every current token position, the head assigns an attention weight to each position that it is allowed to attend to. These weights form an *attention pattern* and sum to one.

It is useful to think of an attention pattern as describing where a head is “looking”. For example, if a head assigns an attention weight of 0.8 to an earlier token and much smaller weights to the remaining positions, most of the information being retrieved by that head at the current position comes from that earlier token.

This interpretation is useful for generating hypotheses about what a head does, but it should not be taken as a complete explanation of the head’s function. Attention weights describe where information is retrieved from; they do not directly tell us whether the retrieved information is subsequently used in a way that matters for the model’s output.

2.2. The induction pattern. Consider a sequence containing

$$\dots A B \dots A.$$

At the second occurrence of A , an induction mechanism should use the earlier occurrence of A to determine what comes next. Since the earlier A was followed by B , the desired prediction is B .

The characteristic attention pattern is therefore slightly more specific than simply attending to the earlier occurrence of A . An induction head should use the current token to find an earlier matching token and then retrieve information about what followed that

earlier token. In the simplified example,

$$\dots A B \dots A \longrightarrow B,$$

the attention at the second A should be directed toward the position containing the earlier B .

This creates a distinctive signature that can be searched for directly in the model’s attention patterns. In this report, this is referred to as the *prefix-matching* or *induction-target* pattern.

The mechanism is particularly interesting because it is a simple example of in-context computation. The model does not need to store a fixed association between A and B in its parameters. Instead, it can use an occurrence of the pair already present in the current context. This provides a concrete example of how information contained in the input can influence later predictions.

2.3. Induction and in-context learning. The induction mechanism is closely related to the broader concept of *in-context learning*. In-context learning refers to the ability of a language model to use information supplied in its input context to improve its predictions without changing its parameters.

The simplest repeated-token induction task is much narrower than the full range of behaviors usually called in-context learning. Nevertheless, it provides a useful mechanistic building block. A model that can detect that a token has occurred before and retrieve what followed it can already perform a simple form of learning from the current context.

Previous work has argued that induction heads are an important component of the mechanisms underlying in-context learning [Ols+22]. In larger or more capable models, the same general idea may also operate over more abstract representations. Instead of requiring the current token to match an earlier token exactly, a model could potentially match related representations and transfer an associated continuation. Spelled out, this means that the model could predict

$$\dots A' B' \dots A \longrightarrow B,$$

where $A' \approx A$ and $B' \approx B$ in some way. This can be viewed as a more general form of sequence copying or an “in-context nearest-neighbor” mechanism. For example, analogous relationships could in principle arise when the model encounters related words or concepts in different languages.

The present experiment does not attempt to establish such an abstract mechanism. It focuses on the simpler and more directly measurable case of exact token repetition.

2.4. Correlation versus causation: ablation. An attention pattern is observational evidence. If a head attends from a repeated token to the token that followed an earlier occurrence, this is evidence that its behavior is consistent with induction. It does not establish that the head is necessary for the model’s induction behavior.

To test causal importance, mechanistic interpretability commonly uses *ablation*. Ablation means deliberately modifying or removing the contribution of a component during a forward pass and measuring how the model’s behavior changes.

In this experiment, a candidate attention head is ablated by replacing its output with either zero or a mean activation computed from a reference set of control sequences. If removing a head decreases the induction score, that provides evidence that the head contributes positively to the measured behavior. Conversely, if removing a head increases the induction score, this suggests that the head may oppose or suppress the behavior.

Ablation does not provide a complete proof of a head’s role either. Neural networks are highly distributed systems, and another component may compensate for the removal

of the ablated component. For this reason, the results in this report are interpreted only as evidence of causal contribution rather than as a complete decomposition of the model’s computation.

3. EXPERIMENTAL SETUP

3.1. Model and implementation. All experiments use GPT-2 Small through TransformerLens. The model has 12 layers and 12 attention heads per layer, giving 144 heads in total. The experiments were run locally on an Apple Silicon MacBook Air using PyTorch’s MPS backend when available, with CPU execution used as a fallback and for numerical validation. The purpose of the implementation was to keep the experiment small enough to run locally while retaining a systematic search over the complete set of attention heads.

3.2. Test sequences. Each test sequence contains 16 tokens divided into two halves of eight tokens. Tokens are sampled uniformly from a pool of 200 GPT-2 vocabulary tokens, excluding special tokens.

The *repeated* condition is constructed by sampling the first eight tokens and copying them exactly into the second half. Thus, if the first half is

$$A_1 \ A_2 \ \dots \ A_8,$$

the complete sequence is

$$A_1 \ A_2 \ \dots \ A_8 \ A_1 \ A_2 \ \dots \ A_8.$$

The *shuffled* condition uses the same basic construction but applies a derangement to the second half, i.e., a permutation in which no element remains in its original position. This preserves the token frequency and sequence length while removing the exact positional repetition present in the repeated condition.

The shuffled sequences therefore serve as a negative control. The model still sees the same type and number of tokens, but the specific repetition structure required for the induction task has been removed.

A total of $N = 200$ repeated sequences and 200 shuffled sequences are used for the main behavioral comparison.

3.3. Induction score. To measure whether the model actually uses the repeated structure, we define an induction logit score. For each position in the second half of a repeated sequence, the target is the token that followed the corresponding token in the first half.

For example, if the sequence contains

$$A \ B \ C \ D \ A \ B \ C \ D,$$

then when evaluating the second A , the induction target is B ; for the second B , the target is C , and so on.

At each predicted position, we calculate the target token’s logit minus the largest logit assigned to any non-target token. A logit is the model’s raw score for a token before the final softmax converts scores into probabilities. A positive margin therefore indicates that the target is preferred over every individual alternative. The final induction score is the average of these margins across predicted positions and then across sequences. This score measures the model’s behavior directly rather than assuming that a particular attention pattern necessarily produces the behavior.

3.4. Attention-pattern screening. We next inspect the attention patterns of all 144 attention heads. For each repeated sequence and each token in the second half, we locate the corresponding token in the first half. We then identify the token that followed that earlier occurrence. The attention weight assigned to this induction target is recorded.

For example, for

$$A\ B\ C\ D\ A\ B\ C\ D,$$

the second A should attend to the earlier B , the second B should attend to the earlier C , and so on.

The score for a head is the mean of these attention weights across the relevant positions and across the 200 repeated sequences. Heads are ranked by this score, and the top 20 are selected for causal ablation.

An earlier version of the experiment instead measured attention to the matching token itself. That quantity can indicate a useful form of token matching, but it is not the complete induction signature because an induction head needs to retrieve information about what followed the matching token. The metric was therefore corrected before the results reported here were produced.

3.5. Causal ablation. The 20 highest-scoring heads from the attention screen are subjected to causal ablation.

The experiment uses two ablation methods. In *zero-ablation*, the output of the selected attention head is replaced by zero. In *mean-ablation*, the output is replaced by the mean activation of that head measured over 100 shuffled reference sequences.

Mean-ablation provides a useful comparison with zero-ablation because zeroing a head can move the network into a substantially different activation regime. Replacing the activation with a typical value from the control distribution is intended to provide a more in-distribution intervention.

For each candidate head, the induction score is measured both before and after ablation. The reported causal effect is defined as the intact induction score minus the ablated induction score. Under this definition, a positive value means that ablating the head decreases induction performance, while a negative value means that ablating the head increases induction performance. In other words, a positive value means that the head contributes positively to induction performance, at least naively.

The ablation experiment uses the first 50 repeated sequences.

3.6. Numerical validation: CPU vs. MPS. All main experiments were run on Apple Silicon using the MPS backend, as indicated by the experiment output. Because numerical differences between hardware backends can potentially affect interpretability experiments, a separate comparison was performed between CPU and MPS execution.

The validation used 20 sequences and the three highest-ranked attention candidates, L5H5, L6H9, and L5H1. The comparison included:

- raw logits;
- attention patterns;
- the induction score;
- mean-ablation reference activations; and
- zero- and mean-ablation scores.

The largest observed absolute difference across the compared quantities was 6.7×10^{-5} in the logits. The induction-score difference was at most 8×10^{-6} , and there were zero argmax mismatches among the 320 compared logit positions.

These differences are substantially smaller than the induction effects reported in the main experiment. The MPS results are therefore treated as numerically equivalent to CPU

execution for the purposes of this experiment, while recognizing that the validation is a sample rather than an exhaustive test of every possible computation.

Quantity	Mean abs. diff.	Max abs. diff.
Logits	3×10^{-6}	6.7×10^{-5}
Attention patterns	$< 10^{-6}$	1.0×10^{-5}
Induction score	3×10^{-6}	8×10^{-6}
Mean-ablation activations	$< 10^{-6}$	1×10^{-6}
Zero-ablation score	4×10^{-6}	1.4×10^{-5}
Mean-ablation score	4×10^{-6}	1.1×10^{-5}

TABLE 1. CPU vs. MPS numerical comparison from `compare_cpu_mps.py`. The comparison uses 20 sequences and the heads L5H5, L6H9, and L5H1 for the ablation measurements. There were zero argmax mismatches out of 320 compared logit positions.

4. RESULTS

4.1. GPT-2 Small exhibits a clear induction effect. The first question is whether the controlled repeated sequences actually produce the expected behavioral effect.

Condition	Mean induction score	Std. dev.
Repeated	2.5717	0.8210
Shuffled	0.0150	0.9756
Difference	2.5568	—

TABLE 2. Induction logit score averaged over $N = 200$ sequences of each condition.

The repeated condition has a mean induction score of 2.5717, while the shuffled control has a mean of only 0.0150. The difference comes out to 2.5568 after rounding.

The result provides strong evidence that the model is exploiting the repetition structure in the constructed sequences. The control condition is particularly useful here because it contains the same general token distribution and sequence length while removing the specific positional correspondence required by the repeated condition.

The relatively large standard deviations show that the effect is not identical across all sequences. Nevertheless, the difference in means is much larger than the variation in the individual sequence scores. The behavioral test therefore establishes a clear target for the mechanistic analysis that follows: the model is performing the induction-like task, and the next question is which internal components implement or influence it.

4.2. Attention-based head ranking. The attention screen ranks all 144 heads according to how strongly they attend to the induction target.

L5H5 has the highest attention score at 0.7939. This means that, on average, approximately 79% of the attention weight assigned by this head at the relevant positions falls on the token identified as the induction target.

The next highest-scoring heads are L6H9, L5H1, and L7H10. Several of these heads have previously been associated with induction behavior in GPT-2 Small, providing an independent point of comparison for the screening procedure.

It is important, however, not to interpret these numbers as causal importance. An attention score of 0.7939 does not mean that L5H5 is “79% responsible” for induction.

Rank	Head	Attention score
1	L5H5	0.7939
2	L6H9	0.7305
3	L5H1	0.7277
4	L7H10	0.7076
5	L10H7	0.5755
6	L7H2	0.5450
7	L10H1	0.4803
8	L5H0	0.4711
9	L9H6	0.4405
10	L11H10	0.4222

TABLE 3. Top 10 of the 20 attention-ranked candidate heads. The full top-20 ranking is stored in the accompanying results file.

It only says that this head’s attention pattern strongly matches the expected induction pattern. The ablation experiment tests the stronger causal question.

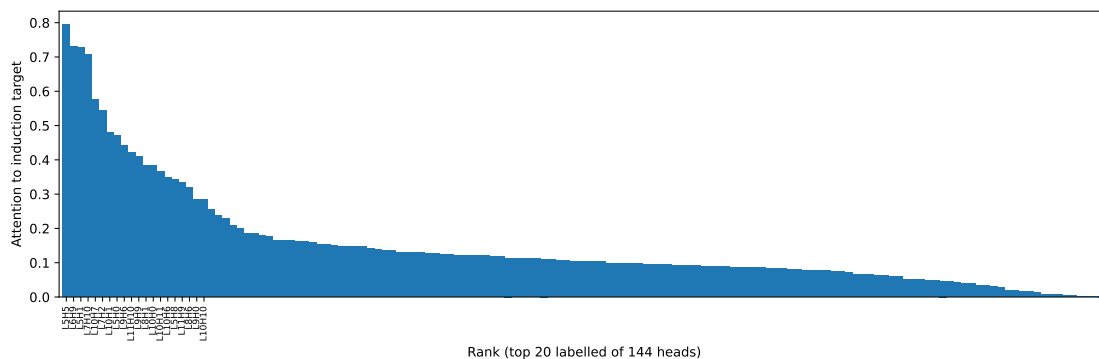


FIGURE 1. Attention-based induction score for all 144 heads, sorted descending. The ranking provides an observational screen for heads whose attention patterns are consistent with induction.

4.3. Causal ablation results. The 20 highest-scoring heads were then subjected to zero- and mean-ablation.

The strongest positive zero-ablation effect belongs to L5H5, with an effect of 0.4446. Several other candidates also show positive effects under both ablation methods, including L5H0, L10H0, L5H1, L7H2, and L6H9. This provides causal evidence that a number of the heads selected by the attention screen contribute positively to the measured induction behavior.

The agreement between the attention screen and the ablation results is particularly informative for L5H5, L5H1, L6H9, and L7H2. These heads both display the expected attention pattern and produce a substantial decrease in induction performance when ablated.

However, the correspondence is not perfect. Some heads with strong attention scores have relatively small or even negative ablation effects. This is exactly the kind of result that would be missed by an attention-only analysis.

Rank	Head	Zero effect ($\pm$ std)	Mean effect ($\pm$ std)
1	L5H5	0.4446 (0.3215)	0.1082 (0.2353)
2	L5H0	0.2404 (0.1606)	0.0505 (0.1413)
3	L10H0	0.2267 (0.1333)	0.2283 (0.1387)
4	L5H1	0.2149 (0.3858)	0.3712 (0.3686)
5	L7H2	0.2071 (0.1385)	0.2255 (0.1374)
6	L6H9	0.1915 (0.3065)	0.2301 (0.2786)
7	L10H6	0.1418 (0.0922)	0.1532 (0.0985)
8	L10H10	0.1186 (0.0673)	0.0801 (0.0695)
9	L9H6	0.1051 (0.1328)	0.0303 (0.1268)
10	L11H9	0.0909 (0.1085)	0.2409 (0.0985)
11	L10H11	0.0815 (0.0780)	0.1103 (0.0787)
12	L8H6	0.0722 (0.1013)	0.0869 (0.0983)
13	L8H1	0.0546 (0.1068)	0.1175 (0.0976)
14	L9H0	0.0347 (0.0444)	0.0061 (0.0421)
15	L10H1	0.0195 (0.1355)	0.2022 (0.1497)
16	L9H9	-0.0048 (0.1086)	0.0444 (0.1016)
17	L5H8	-0.1221 (0.1369)	-0.2691 (0.1574)
18	L7H10	-0.1546 (0.1816)	-0.1054 (0.1530)
19	L11H10	-0.2933 (0.1313)	-0.3100 (0.1288)
20	L10H7	-0.3210 (0.1682)	-0.4947 (0.1780)

TABLE 4. Causal ablation effect on the induction logit score for the 20 attention-ranked candidate heads. Results are reported for $N = 50$ repeated sequences. Positive values indicate that ablation reduces induction performance; negative values indicate that ablation increases it.

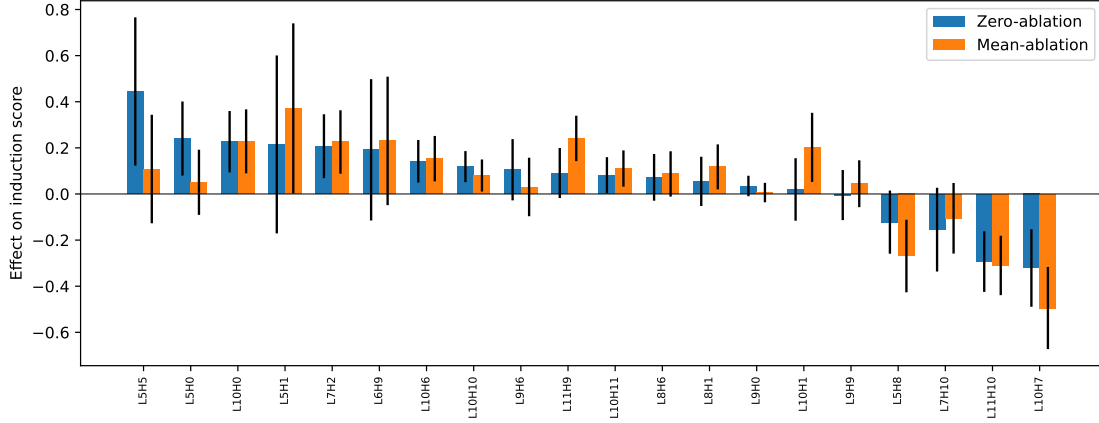


FIGURE 2. Zero- and mean-ablation effects on the induction logit score for the 20 candidate heads, sorted by zero-ablation effect. Positive values indicate that removing the head decreases induction performance.

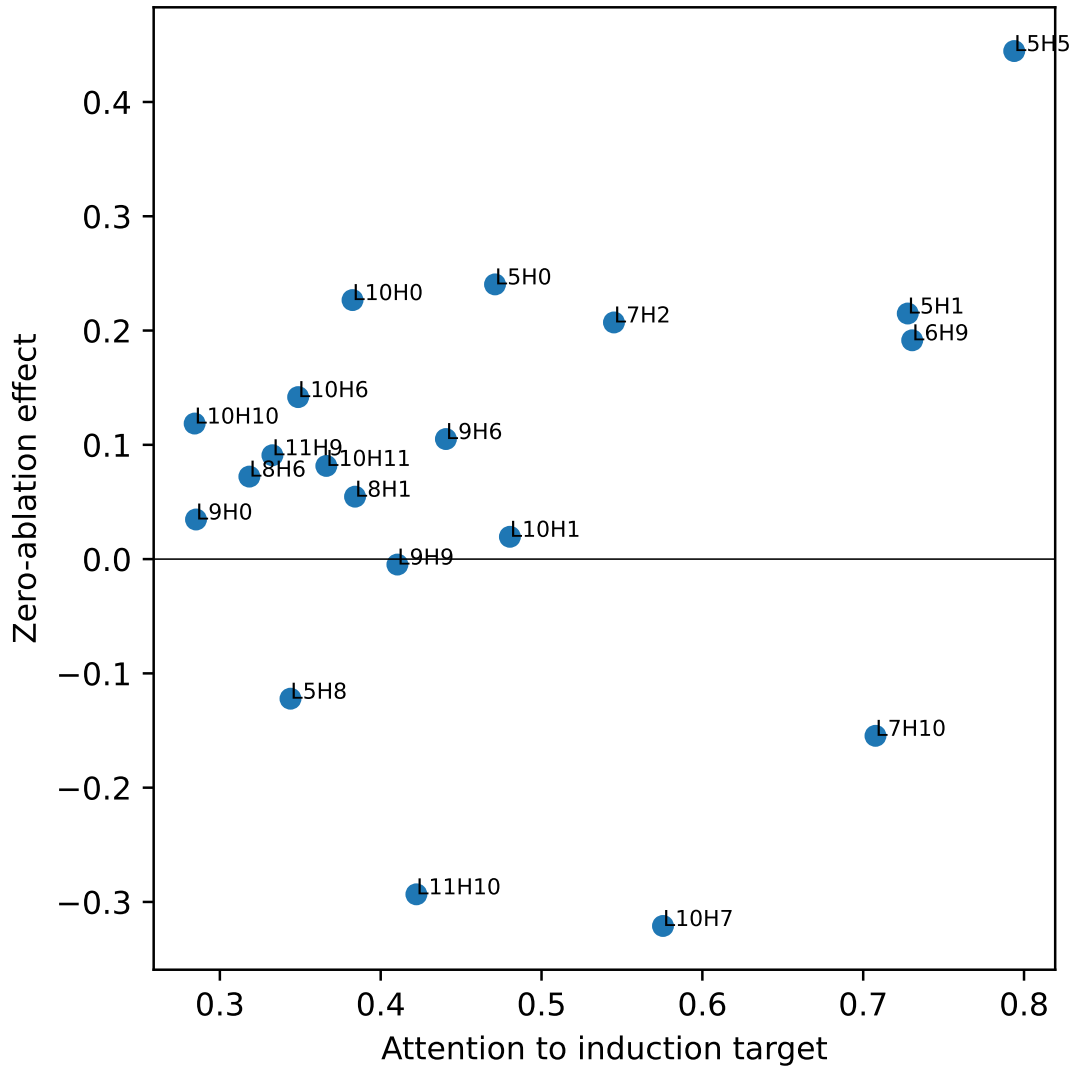


FIGURE 3. Attention score versus zero-ablation effect for each candidate head. The relationship is not perfectly monotonic: strong induction-like attention does not necessarily imply a positive causal contribution. L10H7 is a particularly clear example, combining a relatively high attention score with a negative ablation effect.

5. DISCUSSION

5.1. Agreement with prior work. The attention screen recovers several heads that closely match induction heads previously reported for GPT-2 Small, including L5H5, L6H9, L5H1, L7H10, and L7H2 [Ols+22].

This agreement is useful because the experiment was implemented independently and from scratch. The fact that the simple prefix-matching metric recovers a similar set of heads suggests that the screening procedure is measuring a real property of GPT-2 Small rather than an artifact of a particular example sequence.

At the same time, agreement with previously reported heads should not be treated as proof that the implementation is correct. The more important test is whether the identified heads have a causal effect on the behavior. The ablation results provide that second piece of evidence.

5.2. L5H5 as the strongest positive candidate. L5H5 is the clearest positive result in the experiment. It has the highest attention score, 0.7939, and also produces the largest positive zero-ablation effect, 0.4446.

This agreement between observation and intervention is what we would expect from a head that plays an important role in induction. The attention pattern says that L5H5 is strongly retrieving the information we expect an induction head to retrieve, while the ablation result says that removing its output substantially reduces the measured induction behavior.

The difference between zero- and mean-ablation is nevertheless important. The zero-ablation effect for L5H5 is 0.4446, while the mean-ablation effect is only 0.1082. This suggests that the magnitude of the zero-ablation effect should not be interpreted as a pure measure of induction-specific computation.

Zeroing the head removes its entire output, including any baseline component of that output. Mean-ablation instead replaces it with a typical activation from the control distribution. The smaller mean-ablation effect therefore indicates that part of the large zero-ablation effect may come from the general contribution of the head to the residual stream rather than exclusively from the specific induction computation.

This does not undermine the result. Rather, it shows why multiple intervention baselines are useful when interpreting causal experiments in neural networks.

5.3. A candidate copy-suppression head: L10H7. L10H7 provides the most striking example of why attention and causal importance should be separated.

The head ranks fifth in the attention screen, with an induction-target attention score of 0.5755. Based on the attention pattern alone, it would therefore appear to be a plausible induction candidate.

Its ablation result tells a different story. Zero-ablation produces an effect of -0.3210 , while mean-ablation produces an even more negative effect of -0.4947 . Since the causal effect is defined as intact score minus ablated score, these negative values mean that induction performance actually *increases* when L10H7 is removed.

This is difficult to reconcile with the interpretation that L10H7 is simply an ordinary positive induction head. Instead, the result is consistent with the possibility that the head performs a form of copy-suppression computation: it may attend to information relevant to copying while its overall contribution acts to reduce the final induction signal.

Copy suppression has been discussed previously in analyses of GPT-2 Small [McD+23]. The present experiment does not establish that L10H7 implements exactly the same mechanism described in that work. It does, however, provide a strong motivation for investigating the head further.

The remaining two candidates, L11H10 and L5H8, also have negative effects under both ablation methods. Their effects are smaller than those of L10H7, but they suggest that suppressive or competing components may be a broader part of the induction circuit.

5.4. Why attention alone is not enough. The relationship between attention score and causal effect can be seen directly in Figure 3. If attention alone were a sufficient description of function, we would expect heads with higher induction-target attention to consistently produce larger positive ablation effects.

The experiment does not show such a simple relationship.

Some high-attention heads have strong positive causal effects, while others have weak or negative effects. L10H7 is the clearest example: its attention score is higher than most of the candidate heads, yet removing it improves the measured behavior.

There are several possible explanations. A head may retrieve induction-relevant information but transform or route that information in a way that does not promote the final prediction. It may participate in a competing computation. Alternatively, its contribution may interact with downstream components in a way that is not captured by the attention pattern alone.

The important methodological conclusion is simpler than any particular mechanistic explanation: *attention is evidence about what a head looks at, not a complete description of what the head does.*

5.5. Zero- and mean-ablation disagreement. Several heads show substantial differences between zero- and mean-ablation. L5H1 has an effect of 0.2149 under zero-ablation and 0.3712 under mean-ablation, while L10H1 changes from 0.0195 to 0.2022.

These differences indicate that the measured causal effect depends somewhat on how the intervention is constructed. This is expected in a distributed neural network: replacing a head’s output with zero is a different perturbation from replacing it with a typical activation.

The standard deviations also matter. L5H1, for example, has a zero- ablation standard deviation of 0.3858, which is large relative to its mean effect. The current experiment therefore does not justify a fine-grained interpretation of the exact difference between the two ablation methods for individual heads.

Instead, the most robust conclusions are those where multiple pieces of evidence agree: L5H5 is strongly induction-like and has a substantial positive causal effect, while L10H7 has a strong induction-like attention pattern but a consistently negative causal effect under both ablation methods.

5.6. What the experiment says about the circuit. The results support a picture in which induction in GPT-2 Small is not implemented by a single isolated “induction head”. Several heads appear to contribute positively, while others may have competing or suppressive effects.

This is consistent with the broader idea of a *circuit*: a behavior can emerge from interactions between multiple components rather than from a single component acting independently.

The present experiment does not yet identify the complete circuit. It primarily establishes candidate components and their causal effects. The next step would be to investigate how these heads interact, for example by tracing their inputs and outputs through the residual stream or by using activation patching to determine which information is necessary at different stages of the computation.

6. LIMITATIONS

The experiment is intentionally small and therefore has several limitations.

- **Different sample sizes.** The behavioral existence test uses $N = 200$ sequences, while the ablation experiment uses $N = 50$. The ablation estimates therefore have substantially more sampling noise, as reflected in several of the standard deviations in Table 4.
- **Single-head ablation.** Ablating one head at a time cannot distinguish between a head having no causal role and a head whose role is redundant or compensated by other components. A stronger circuit analysis would test combinations of heads and investigate possible redundancy or “self-repair”.
- **Random token selection.** The token pool is sampled from the GPT-2 vocabulary. This makes the experiment easy to generate and reduces semantic assumptions, but it also allows unusual, rare, or otherwise atypical tokens to appear. Future experiments could use a more controlled token set.
- **Limited behavioral metric.** The induction score is a logit-margin measure designed specifically for this task. It captures whether the target token is preferred over alternatives, but it is not a complete description of the model’s behavior.
- **No formal statistical testing.** The report primarily uses means and standard deviations. A more rigorous analysis would use paired statistical tests or bootstrap confidence intervals across the same sequences.
- **Limited MPS validation.** CPU/MPS numerical parity was checked on 20 sequences and three heads rather than across the complete main experiment. The results provide evidence of numerical agreement but do not constitute an exhaustive hardware-validation study. This, however, is not very relevant for the larger experiment.
- **Attention-based candidate selection.** Only the top 20 heads from the attention screen were ablated. Consequently, a head with a weak induction-target attention score but an important causal role could have been missed.
- **No complete circuit reconstruction.** The experiment identifies candidate heads and tests their individual effects, but it does not yet establish the full sequence of computations connecting token matching to the final output logits.

7. CONCLUSION

This experiment provides a small but concrete demonstration of mechanistic interpretability in GPT-2 Small. The model shows a clear induction effect on controlled repeated-token sequences, with a mean induction score of 2.5717 compared with 0.0150 for shuffled controls. An attention-based screen over all 144 heads then recovers several heads whose behavior closely matches induction heads reported in previous work.

Causal ablation adds an important second layer of evidence. L5H5 is the strongest positive candidate, combining the highest induction-target attention score with the largest positive zero-ablation effect. Other screened heads also show positive causal effects. At the same time, L10H7 demonstrates that an induction-like attention pattern does not necessarily imply a positive causal role: ablating the head actually improves induction performance, making it a plausible candidate for a suppressive or competing computation.

The main lesson is therefore methodological as much as mechanistic. Attention patterns provide a useful way to generate hypotheses about what transformer components are doing, but causal interventions are needed to determine whether those components actually matter for the behavior of interest. The present results establish several promising components of the GPT-2 Small induction circuit, but a complete mechanistic account will require further

experiments on interactions between these heads and the information flowing through the residual stream.

8. REPRODUCIBILITY

All experiments use a fixed random seed, `SEED = 42`. The main experiment is implemented in `experiments/full_induction_experiment.py`, while numerical CPU/MPS validation is implemented in `experiments/compare_cpu_mps.py`.

The main experiment uses 200 repeated sequences and 200 shuffled controls for the behavioral and attention analyses. The top 20 attention-ranked heads are evaluated using causal ablation on 50 repeated sequences. Mean-ablation reference activations are computed from 100 shuffled sequences.

Raw attention-ranking results are stored in `results/attention_head_scores.csv`, and causal ablation results are stored in `results/head_ablation_results.csv`.

The experiments are designed to run locally using GPT-2 Small and TransformerLens, with MPS support on Apple Silicon and CPU fallback. The fixed random seed and controlled sequence-generation procedure are intended to make the reported results reproducible.

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